

Hints & Solutions

2. Work done by friction is zero, since point of application remains at rest.
3. Always zero as $\vec{F}_{cp} \perp \vec{V}$ so $\theta = \pi/2$
So work done = 0
4. $F_c = \frac{-dU}{dx}$
- $$\int_{x_i}^{x_f} F_c dx = - \int_{U_i}^{U_f} dU$$
- Work done by $F_c = -\Delta U$
5. Work done by kinetic friction can be +ve/-ve/zero.
6. $E = K + U = \text{constant}$
K and U can change internally.
So if U increase then K decrease
if U decrease then K increase
if U remains constant, K remains constant

13. Area under P – x curve = 30

$$\int P dx = 30$$

$$\int \left(mv \frac{dv}{dx} \right) v dx = 30$$

$$\int_1^v mv^2 dv = 30$$

$$\frac{10}{7} \left(\frac{v^3}{3} \right)_1^v = 30 \Rightarrow v = 4 \text{ m/s}$$

14. Total work done by static friction on a complete system is always zero and total work done by kinetic friction on a complete system is always negative.
16. During the short time of collision, some K.E. gets converted to potential energy.